USED PAPERS AUG/SEP-2021

Code No: 156DX **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2021
NON-CONVENTIONAL ENERGY SOURCES
(Common to EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the solar radiation on tilted surface.
b) Write a note on extraterrestrial and terrestrial solar radiation. [7+8]

2.a) What are the various instruments used for solar radiation measurement? Explain the working of Pyranometer with a neat sketch.
b) What is the need for alternate energy sources? Explain by considering solar energy. [8+7]

3.a) Explain the carnot cycle and its limitations.
b) Calculate the monthly average hourly radiation falling on a flat plate collector facing south ($\gamma = 0$) with a slope of 15^0, given the following data:
Location : Chennai (13^0 00' N) ;
Month : October ;
Time : 1100 to 1200 (LAT) ;
$I_g = 2408$ kJ/m^2 – h ; $I_d = 1073$ kJ/m^2 – h.
Assume the ground reflectivity to be 0.2. [8+7]

4.a) What is the principle of a fuel cell? Explain the main components of fuel cell and their functions.
b) Define and explain the figure of merit of a thermoelectric device. [9+6]

5.a) Describe the solar pond used for solar energy conversion and storage.
b) Sketch and explain the single basin type tidal power plant operation. [8+7]

6.a) Compare and contrast between the different types of thermodynamic cycles used for OTEC.
b) Discuss the economic analysis of a mini hydel power plant. [8+7]

7.a) Explain the following operating characteristics of wind mills: Cut-out speed, Betz limit.
b) Explain briefly about the vertical axis wind mills with neat sketches. [6+9]

8.a) Distinguish between Anaerobic digestion and Aerobic digestion.
b) Explain the main types of turbines that may be used for Geothermal energy conversion. [7+8]

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